

EE1101: Circuits and Network Analysis
Assignment - 14

Handed out: 16 - Nov - 2024

Due : 28 - Nov - 2024 (before 5 PM)

Instructions :

1. Please upload your assignment solutions to the course page on the Canvas platform. Only solutions submitted through canvas will be reviewed. For specific guidelines, refer to the instructions provided on the course page.
2. It is suggested that you attempt all the problems. However, it is sufficient to submit solutions for problems that total 10 points.
3. Submissions received after the deadline will attract negative marking. Ensure that your submissions are named in the following format: RollNo-Assignment-14.pdf.

1. (4 points) Consider two EE cores stacked together as shown in Fig. 1. Compute the inductance of the configuration considering N turns are wound on the central limb. Given the mean lengths of each path (as shown in Fig. 1), compute the equivalent magnetic length of the entire configuration. Consider the area of the cross section to be A_c and the permeability of the core to be μ_c .

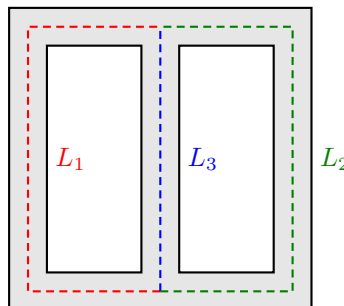


Fig. 1: Magnetic configuration with two EE Cores

2. (8 points) Consider a EE core E 42/21/9 that has a core cross sectional area of 1.07 cm^2 , relative permeability of 2100 (can vary within $\pm 25\%$ around the nominal value) and mean magnetic length of 108.5 mm.
 - (a) (2 points) Compute the number of turns required to synthesize an inductor whose inductance is 0.4 mH.
 - (b) (2 points) Compute the variation in inductance that can result from variation in relative permeability (use $\pm 25\%$ as the variation limit).
 - (c) (2 points) If the saturation flux density (B_{sat}) of the core is 1T, compute the current at which the chosen core can saturate.
 - (d) (2 points) Compute the flux linking the winding under saturation.

3. (5 points) An inductor is designed of the form shown in Fig. 2. The area of cross section is 3.6 cm^2 and the mean cross length of the core is 15 cm. The permeability of the core to be $2100\mu_0$ and the saturation flux density of the core is 1.0 T.

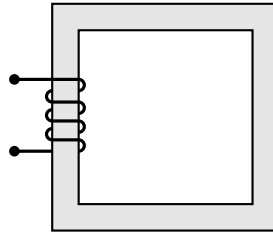


Fig. 2: A simple inductor with rectangular core

- (a) (2 points) Compute the number of turns needed to synthesize an inductance of 6 mH.
- (b) (2 points) Compute the maximum current that the winding can carry before the core saturates.
- (c) (1 point) Compute the net flux linking the winding under saturation (λ_{sat}).
4. (5 points) Consider the same magnetic core as in Problem 3. An air gap of 1 cm is introduced in the core with all other parameters remaining fixed.
- (a) (2 points) Compute the number of turns needed to synthesize an inductance of 6 mH.
- (b) (2 points) Compute the maximum current that the winding can carry before the core saturates.
- (c) (1 point) Compute the net flux linking the winding under saturation (λ_{sat}).